

Session 20: Getting More from OpenDaylight — Detailed Explanation

1. OpenDaylight and AAA (Authentication, Authorization, and Accounting)

- AAA is a framework to control access to network resources.
- ODL integrates AAA mechanisms to authenticate users or applications accessing the controller.
- Ensures **security** by verifying identity (Authentication), determining permissions (Authorization), and logging actions (Accounting).
- Typically, AAA is implemented via protocols like **RADIUS**, **TACACS+**, or **LDAP** integrated with ODL's northbound interfaces.
- Helps prevent unauthorized access to sensitive SDN controller functions.

2. Introduction to OVSDB Virtualization

- OVSDB (Open vSwitch Database Protocol) is a management protocol for Open vSwitch.

- It manages switch configuration and state data, such as ports, interfaces, QoS, and tunnels.
- **OVSDB Virtualization** involves using OVSDB to create virtual network elements (e.g., virtual switches, bridges) on a single physical host.
- This supports multi-tenant isolation and dynamic network provisioning.
- ODL acts as a controller that communicates with OVSDB servers on switches to configure virtualized network topologies.

3. Application Intents and Group Based Policy

- **Application Intents** in ODL allow developers to define *what* the network should do, without specifying *how*.
- Intents abstract low-level configurations, letting ODL translate them into concrete network policies.
- **Group Based Policy (GBP)** is a model that defines network behavior based on groups of endpoints (e.g., users, VMs).

- It simplifies security and network policy management by grouping endpoints and applying policies between groups.
- ODL's GBP project integrates intent-based networking and policy enforcement.

4. Service Function Chaining (SFC)

- **Service Function Chaining** is the process of steering traffic through a sequence of network services (firewall, DPI, NAT, etc.).
- ODL supports SFC by allowing dynamic chaining of virtual or physical network functions.
- It uses protocols like **Network Service Headers (NSH)** to encapsulate traffic and track chains.
- Enables flexible and programmable insertion of services in the data path.

5. LISP Flow Mapping

- LISP (Locator/ID Separation Protocol) separates device identity (ID) from location (Locator).
- Flow mapping helps route packets by mapping endpoint IDs to current locations.
- ODL can act as a control plane component managing LISP mappings dynamically.
- Useful in mobility, multi-homing, and scalable network architectures.

6. Virtual Tenant Networks (VTN)

- VTN provides virtualized Layer 2/Layer 3 network slices for different tenants over a shared physical infrastructure.
- ODL's VTN project enables multi-tenancy by isolating tenant traffic and resources.
- It supports creating virtual routers, switches, and links, ensuring tenant isolation.
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Helps cloud providers or enterprises efficiently manage multi-tenant environments.